PlacementIQ Enterprise

AI-Powered Placement and Career Readiness System

Team Members:

Shaganti Vivek Goud	23EG105F59
Puram Preethi	23EG105F52
Akshaya Reddy	23EG105F03
Bandari laxmisri	23EG105F09

Guided by:

Dr. Kamalakar Ramineni

Assistant Professor

ABSTRACT

- PlacementIQ Enterprise centralizes campus drives, company details, eligibility, packages, and alumni.
- It analyzes CGPA, skills, resume, projects, internships, and coding profiles to assess placement readiness.
- AI predicts Eligible, Nearly Eligible, Can Become Eligible, and Not Eligible companies while identifying skill gaps.
- Provides Company Match Score, Placement Readiness, Salary Prediction, Interview Readiness, and personalized roadmaps.

INTRODUCTION

- Campus placement is the most important milestone for engineering students, but drive details reach them through portals, spreadsheets, e-mail and WhatsApp groups.
- Students rarely know which companies they are eligible for, why they fall short, or which skill to improve first.
- PlacementIQ Enterprise is an AI-powered placement platform: each college onboards its own students, posts drives and monitors readiness by year, branch and section.
- For every student and company it gives an explainable category (Eligible, Nearly Eligible, Can Become Eligible, Not Eligible) with the exact gaps and a personalised plan.

66 recruiters scored

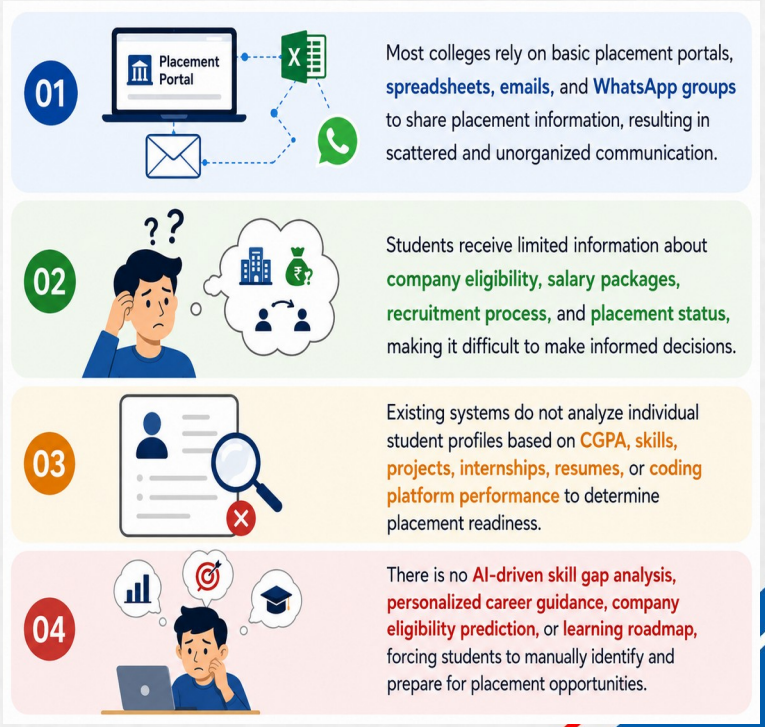
4 eligibility categories

4 user roles

8 skill areas tracked

PROBLEM STATEMENT

- Students have limited information about campus placement opportunities, company eligibility, salary packages, and recruitment processes.
- Existing systems do not evaluate individual placement readiness based on academics, skills, projects, internships, and coding profiles.
- Students cannot easily identify eligible companies, skill gaps, or the improvements required to secure desired placements.
- An AI-powered Placement Operating System is needed to provide company recommendations, skill-gap analysis, personalized roadmaps, alumni mentorship, and real-time placement insights.



EXISTING SYSTEM

Current placement process is scattered, manual and information-limited

CURRENT PRACTICE

- Placement Portals
- Spreadsheets (Excel)
- Emails
- WhatsApp Groups

Information is scattered across multiple platforms and tools

01



SCATTERED COMMUNICATION

Most colleges rely on basic placement portals, spreadsheets, emails, and WhatsApp groups to share placement information, resulting in scattered and unorganized communication.

02



LIMITED INFORMATION

Students receive limited information about company eligibility, salary packages, recruitment process, and placement status, making it difficult to make informed decisions.

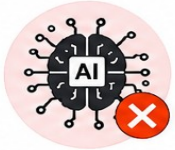
03



LIMITED STUDENT PROFILE ANALYSIS

Existing systems do not analyze individual student profiles based on CGPA, skills, projects, internships, resumes, or coding platform performance to determine placement readiness.

04



NO AI-DRIVEN GUIDANCE OR PREDICTION

There is no AI-driven skill gap analysis, personalized career guidance, company eligibility prediction, or learning roadmap, forcing students to manually identify and prepare for placement opportunities.

IMPACT ON STUDENTS



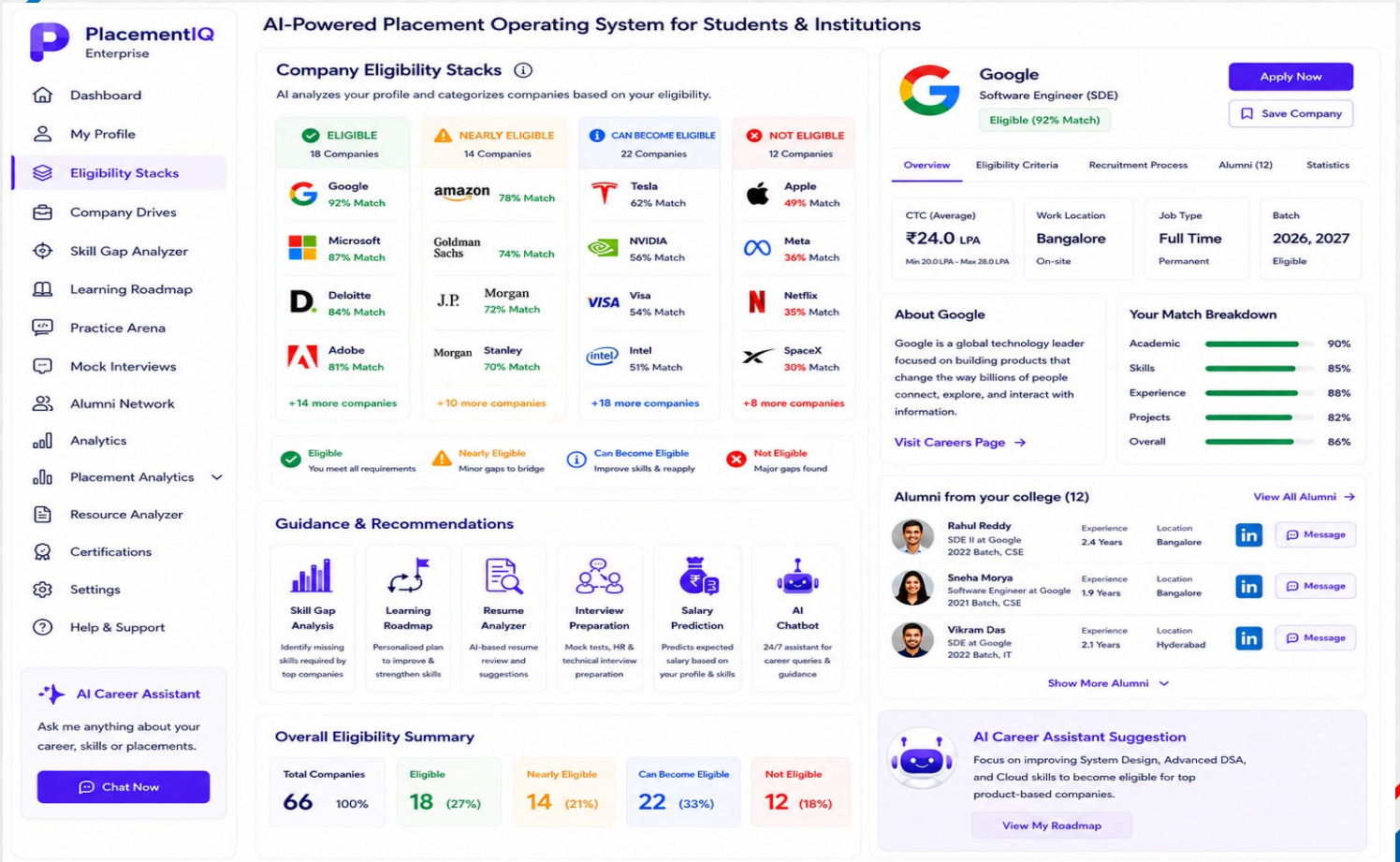
- Unclear about eligibility for companies
- Lack of proper guidance and planning
- Missed opportunities
- Unprepared for placement process

LOW PLACEMENT OUTCOMES

The existing system is manual, unorganized, and limited in scope, which leads to poor decision-making and low placement readiness.

PROPOSED SYSTEM

ANURAG UNIVERSITY



LITERATURE SURVEY

Year	Publisher	Author	Title	Method	Merits	Demerits
2025	IEEE (ICISS)	F. B. Kunwar et al.	Enhanced Student Placement Prediction and Skill Recommendation System	CNN-BiLSTM with attention	Predicts placement and recommends skills	Black-box model; no per-company reason
2024	ACM Computing Surveys	Y. Mashayekhi et al.	A Challenge-based Survey of E-recruitment Recommendation Systems	Survey of job and candidate recommenders	Maps open challenges: sparsity, fairness, explainability	Recruiter-centric; no student guidance
2023	Springer (IJAIED)	A. José-García et al.	C3-IoC: A Career Guidance System for Assessing Student Skills	ML skill similarity + job-role network	Links student skills to IT job roles	Not tied to campus drives or eligibility rules
2023	IEEE (CSITSS)	M. S. A. Basha et al.	Unraveling Campus Placement Success with Predictive ML Models	EDA + supervised classifiers	Shows academic factors predict placement	Only placed or not placed; no improvement plan
2023	IEEE (UBMK)	S. Tanberk et al.	Resume Matching Framework via Ranking and Sorting Using NLP and Deep Learning	NLP + BERT resume ranking	Semantic resume and JD matching	Ranks for recruiters; no feedback to students
2021	IEEE Access	R. Alamri, B. Alharbi	Explainable Student Performance Prediction Models: A Systematic Review	Systematic literature review	Catalogues explainability techniques	Few deployed explainable models found

SOFTWARE AND HARDWARE REQUIREMENTS

Software Requirements

- Frontend: React 18, TypeScript, Vite 5, Tailwind CSS, Recharts
- Backend: Supabase (PostgreSQL, Auth, Storage, row-level security)
- Serverless API: Vercel Edge Functions (Node.js runtime)
- AI: Groq LLM API (gpt-oss-120b, fallback gpt-oss-20b)
- Parsing: pdf.js (PDF) and mammoth (DOCX) resumes
- Integrations: GitHub, LeetCode, Codeforces, CodeChef
- Tools: VS Code, Git and GitHub, Node.js 20+, npm
- Client: latest Chrome, Edge, Firefox or Safari

Hardware Requirements

- Developer PC: Intel i5 / AMD Ryzen 5 / Apple M1 or better
- RAM: 8 GB minimum (16 GB recommended)
- Storage: 10 GB free (SSD preferred)
- Internet: stable broadband for cloud services
- End users: any laptop or phone with a modern browser (4 GB RAM, 360 px+ screen)
- Server: none on premises; fully managed cloud (Supabase + Vercel)

MODULES

Student Module

- Profile, resume and coding links
- Eligibility stacks for 66 companies
- Skill gaps, roadmap, practice
- What-If simulator, AI assistant
- Apply to campus drives

Placement Cell Module

- Roster upload with sections
- Year / branch / section monitoring
- Student rankings and filters
- Post drives, manage applicants
- Add recruiters, view feedback

Recruiter Module

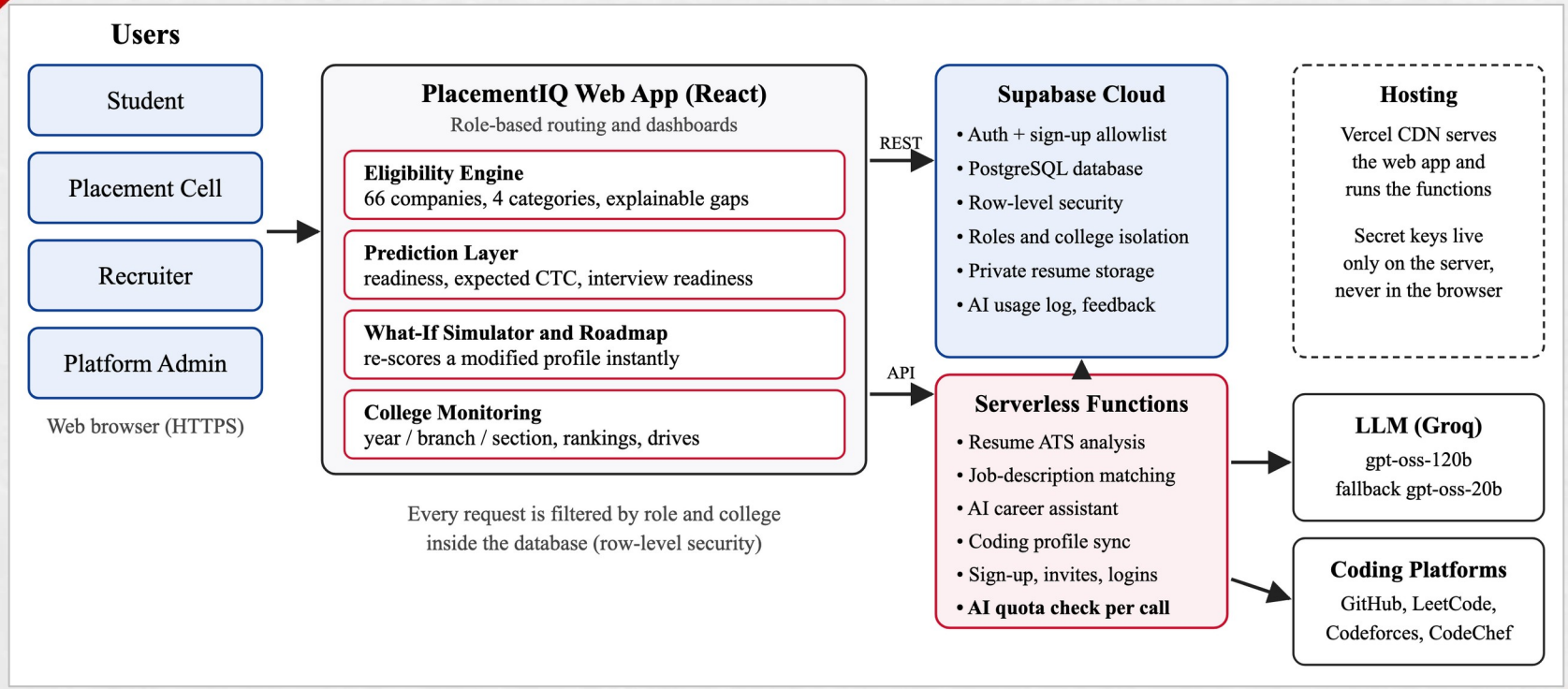
- Post drives for own company
- Ranked applicants with match %
- Full profiles of applicants only
- Shortlist, interview, offer

Platform Admin Module

- Create colleges and logins
- Plans, seats and pricing
- Usage, storage and AI cost
- Profit and margin view

Core engines: Eligibility Engine | Prediction Layer (readiness, expected CTC, interview readiness) | Resume and JD Analyzer (LLM) | Access Control (row-level security)

SYSTEM ARCHITECTURE



IMPLEMENTATION / METHODOLOGY

ANURAG UNIVERSITY

1. College Onboarding

- Platform admin creates the college; placement cell uploads the roster (e-mail, roll no., branch, year, section).

2. Student Onboarding

- Rostered students sign up, complete profile, upload resume and link coding profiles.

3. Data Processing

- Resume parsed and scored (6 ATS checks via LLM); GitHub, LeetCode, Codeforces, CodeChef synced.

4. Eligibility Engine

- Profile scored against 66 recruiters and college drives; 4 categories with exact gaps.

5. Predictions & Guidance

- Readiness, expected CTC, interview readiness, What-If, roadmap and AI assistant.

6. Placement Actions

- Cell posts drives, ranks students, monitors classes; recruiters shortlist; feedback loop.

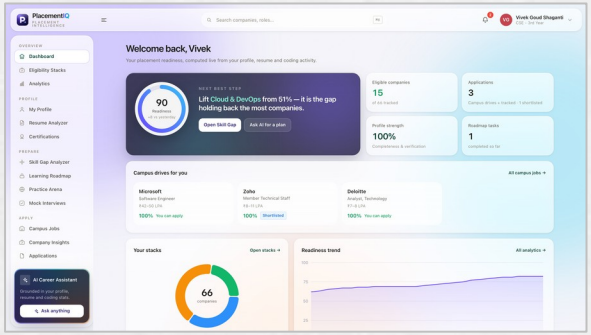
ALGORITHM / METHODOLOGY

1. Input: CGPA, backlogs, Class X/XII %, 8 skill levels, projects, internships, resume and coding statistics.
2. Compute academic score A, skill score S (every point below a required level costs 1.2), experience E and projects P.
3. Match score M = 0.15A + 0.60S + 0.10E + 0.15P for each of the 66 companies.
4. Assign category: Eligible (M ≥ 88, largest gap ≤ 5), Nearly (M ≥ 75, gap ≤ 15), Can Become (M ≥ 55 or near academic cut-off), else Not Eligible.
5. Explain every decision with unmet criteria and gaps; compute readiness, expected CTC and interview readiness.
6. What-If: re-run the same steps on a modified profile and show which companies move up.

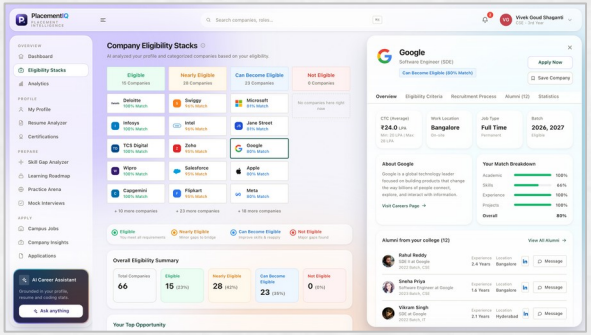
Key Formulas

- $A = 25\,[\min(1, g/g^*) + \min(1, X/X^*) + \min(1, XII/XII^*) + b]$
- $S = 100 - 1.2\,\Sigma \max(0, req - level)$
- $M = 0.15A + 0.60S + 0.10E + 0.15P$
- Readiness = mean M over target companies
- Expected CTC = $\Sigma\, w \cdot CTC / \Sigma\, w$, w = category weight × M
- Interview = 0.3 DSA + 0.15 Apt + 0.15 SD + 0.2 Mock + 0.1 Resume + 0.1 Exp

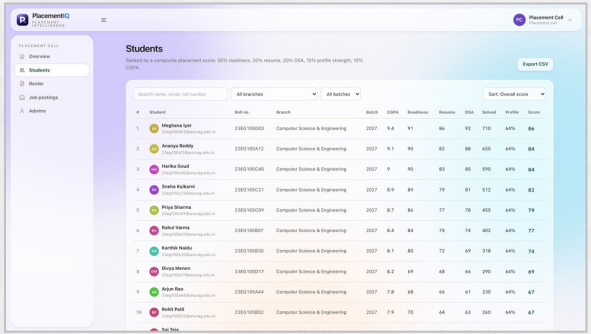
IMPLEMENTATION & SCREENSHOTS



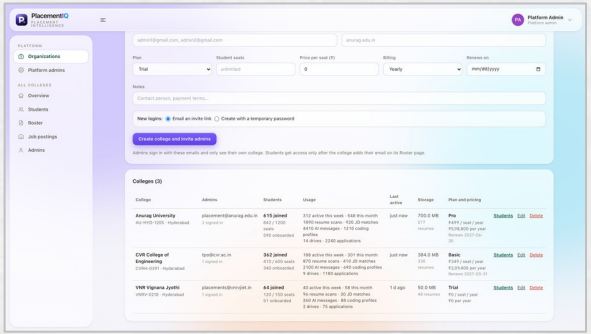
Student dashboard



Eligibility stacks + company panel

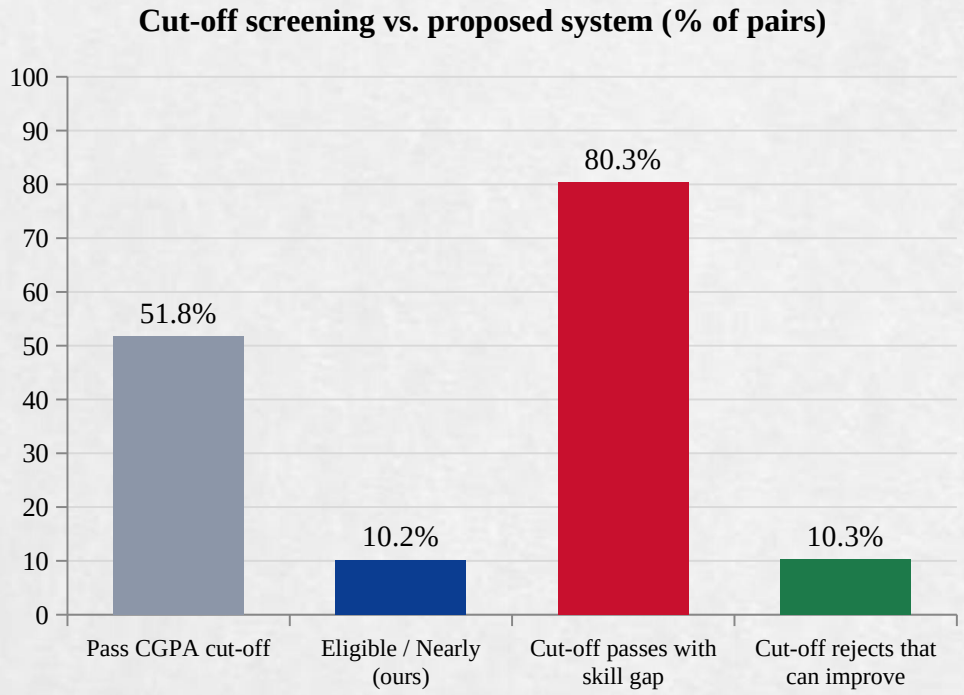


Placement cell: ranked students



Platform admin: colleges and usage

RESULTS / OUTPUT



0.18 ms
to score one student against all 66 companies

100%
of 629,068 non-eligible decisions explained

0
monotonicity violations in 594,000 checks

19 / 19
access-control tests passed on the live database

TESTING

ID	Test Case	Type	Expected / Observed	Result
T1	Sign-up with an unregistered e-mail	Security	Account creation rejected	Pass
T2	Student sign-up with wrong roll number	Security	Rejected with clear message	Pass
T3	Placement cell reads another college	Security	Only own college visible	Pass
T4	Recruiter posts for another company	Security	Blocked by database policy	Pass
T5	AI use beyond plan quota	Functional	Action refused after limit	Pass
T6	Reason for every non-eligible result	Unit / property	629,068 of 629,068 explained	Pass
T7	Improving a skill never lowers a score	Property	0 violations in 594,000 checks	Pass
T8	Scoring speed per student	Performance	< 1 ms (measured 0.18 ms)	Pass
T9	Role routing, What-If, filters on desktop and mobile	Functional (UI)	Correct pages and results	Pass

Security tests T1-T5 run against the live database inside a rolled-back transaction (19 scenarios, all passed).

CONCLUSION & FUTURE SCOPE

Conclusion

- PlacementIQ replaces CGPA cut-off lists and scattered messages with explainable, per-company eligibility for every student.
- Scores 66 companies in 0.18 ms, explains every non-eligible decision and never penalises improvement.
- Revealed that 80.3% of cut-off-eligible pairs hide a major skill gap.
- Role-based access keeps every college and recruiter isolated (19/19 tests).

Future Scope

- Pilot across all batches at Anurag University and train a learned model on real outcomes.
- Interview scheduling, offer tracking and faculty mentor roles.
- Mobile app, single sign-on and multi-college analytics.
- Scale on AWS with fairness audits and DPDP Act 2023 compliance.

REFERENCES

1. F. B. Kunwar et al., "Enhanced student placement prediction and skill recommendation system using CNN-BiLSTM with attention networks," Proc. IEEE ICISS, 2025, pp. 1450-1456.
2. Y. Mashayekhi, N. Li, B. Kang, J. Lijffijt, and T. De Bie, "A challenge-based survey of e-recruitment recommendation systems," ACM Comput. Surv., vol. 56, no. 10, 2024.
3. A. José-García et al., "C3-IoC: A career guidance system for assessing student skills using machine learning and network visualisation," Int. J. Artif. Intell. Educ., vol. 33, 2023.
4. M. S. A. Basha et al., "Unraveling campus placement success integrating exploratory insights with predictive machine learning models," Proc. IEEE CSITSS, 2023.
5. S. Tanberk et al., "Resume matching framework via ranking and sorting using NLP and deep learning," Proc. IEEE UBMK, 2023, pp. 453-458.
6. R. Alamri and B. Alharbi, "Explainable student performance prediction models: A systematic review," IEEE Access, vol. 9, pp. 33 132-33 143, 2021.
7. C. Qin et al., "Enhancing person-job fit for talent recruitment: An ability-aware neural network approach," Proc. ACM SIGIR, 2018, pp. 25-34.
8. R. S. Sandhu et al., "Role-based access control models," IEEE Computer, vol. 29, no. 2, pp. 38-47, 1996.
9. OpenAI, "gpt-oss-120b & gpt-oss-20b model card," arXiv:2508.10925, 2025.

Queries?